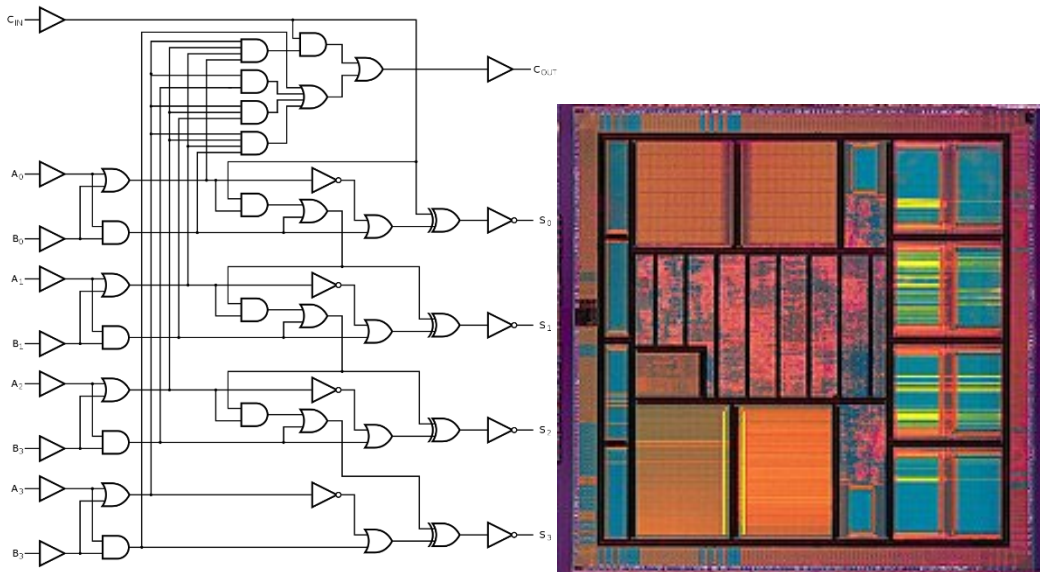


Advanced VLSI Project





Final Project:

In the final project you will need to build a full implementation design of RISC-V. You will start with RTL and need to finish in full Place and Route design.

Your steps are:

- Build Lib (ndm).
- Read RTL
- Synthesis
- Place (including Power nets)
- Clock Tree
- Route
- Add filler cells
- Check Final status

Build Lib:

Technology file:

/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_TECH_DATA/tf/saed14nm_1p9m.tf

Reference Libraries:

/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_STD_LVT/ndm/saed14lvt_base_frame_timing.ndm
/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_STD_RVT/ndm/saed14rvt_base_frame_timing.ndm
/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_STD_HVT/ndm/saed14hvt_base_frame_timing.ndm

Parasitic files:

/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_TECH_DATA/tlup/saed14nm_1p9m_Cmax.tlup
/tech/synopsys/lib/SAED14_EDK/SAED14nm_EDK_TECH_DATA/tlup/saed14nm_1p9m_Cmin.tlup

RTL files:

riscv_core_all.v riscv_defs.v

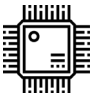
(Both can be found in directory /tech/synopsys/Adv_VLSI_2026/riscv_rtl)

Constraints File

riscv.sdc

You also get some scripts that will help you run the design:

1. Power nets - create_pg_network.tcl
2. Route rules - route.tcl
3. Add boundary cells - insert_boundary_and_tap_cells.tcl
4. Add filler cells - filler.tcl



All the scripts are under:
/tech/synopsys/Adv_VLSI_2026

Reports:

1. Location of the final Fusion Compiler run's.
 - a. /project/advvlsi/users/<user_name>/ws/project_2026
2. Final scripts
 - a. Can be in the directory of your project, what is the file name?
3. Clock Skew report
 - a. You can find the command in the CTS lecture.
4. Final timing report
 - a. Can be in the directory of your project, what is the file name?
5. Final PG (Power/ Ground) DRC report
 - a. Can be in the directory of your project, what is the file name?

Final reports:

When you finish the project send email to webb@tauex.tau.ac.il

The email should include:

- the team names and ID.
- Location of the final Fusion Compile.

Good luck,

Zvika, Adi, Inbal.